

# ICT 1404 Maths & Statistics for Computing

## Probability

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The *probability* of an event is the chance or likelihood of the events occurring.

If A is an event the Probability is denoted by  $P(A)$

$$0 \leq P(A) \leq 1$$

## Axioms of Probability

- The probability of each outcome is greater than or equal to zero.  $0 \leq P(A)$
- The probability of a certain or sure event is one.  $P(A)=1$
- The probability of an impossible event is zero.  $P(A)=0$
- Therefore, probability of an event is always between zero and one.

$$0 \leq P(A) \leq 1$$

## Examples:

- Getting 'Head' when a coin is tossed
- Obtaining 5% of defected items out of all items
- Occurring an accident at a specific place of a road
- Completing the construction of Hostel building
- Receiving an order of goods on time.
- Sun rises at tomorrow
- Awarding a First Class in your Degree.

## Methods of Approach to probability

1. Classical Approach
2. Relative Frequency Approach
3. Personalistic Approach
4. Mathematical approach

# 1. Classical Approach

In classical approach each and every events are considered as having equal chance to happen (equally likely events).

$$P(A)=n(A)/n(S)$$

example1: Getting 'head ' when a coin is tossed.

Getting 'head ' - H

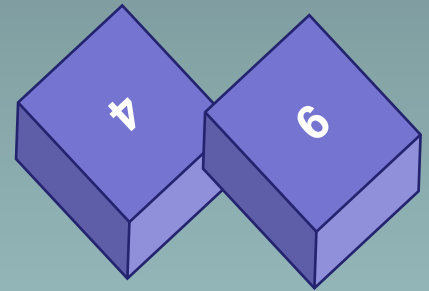
$$Pr(H)=n(H)/n(S)$$

Getting 'tail' - T

$$Pr(H)=1/2$$

$$S=\{ H, T \} \quad n(S)=2 \quad n(H)=1$$

Example 2: Getting sum of the values is ten when two dice are thrown.



$S = \{ (1,1), (1,2), \dots, (1,6), (2,1), (2,2), \dots, (6,1), \dots, (6,6) \}$

$A = \{ (4,6), (5,5), (6,4) \}$

$n(S)=36 \quad n(A)=3$

$\Pr(A)=3/36 = \mathbf{1/12}$

# 1. Relative frequency Approach

If outcomes are not with equal chances then classical approach cannot be used. Then it is used Relative frequency Approach.

In this approach number the occurrences of the event is divided by total number of times random experiment done.

$$\text{Probability} = \frac{\text{number of times event occurred}}{\text{Total Number of time experiment done}}$$

## Example: Tossing a coin

| No. of times experiment done | No. of times getting Head | P(A)    |
|------------------------------|---------------------------|---------|
| 10                           | 4                         | 0.4     |
| 100                          | 38                        | 0.38    |
| 1,000                        | 381                       | 0.381   |
| 10,000                       | 3805                      | 0.3805  |
| 100,000                      | 38015                     | 0.38015 |

Higher the number of times experiment done higher accuracy of probability.  
Therefore  $P(A)=0.38$

# Random Experiment

A process which generate outcomes with uncertainty.

Examples:

Tossing a coin

Throwing a die

Getting a card from a card pack

## Sample space

The set of all possible outcomes within the random experiment.

Sample space can be represent in following ways.

Sets

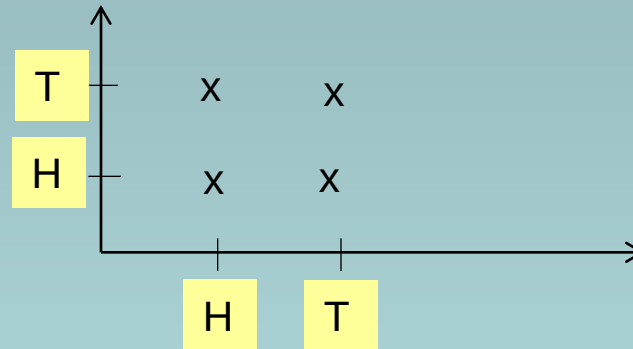
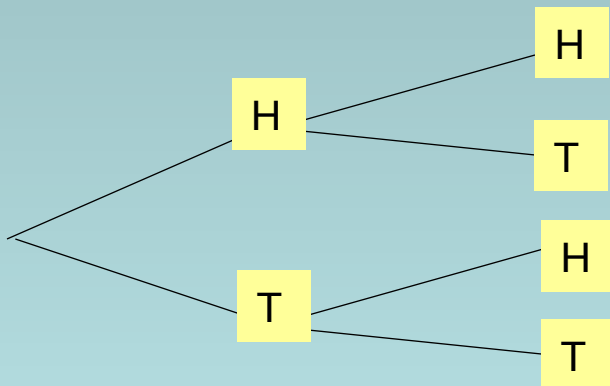
Tree diagrams

Graphical method

Example:

Tossing a fair coin

$S = \{HH, HT, TH, TT\}$



## event space

Collection of possible outcomes  
relevant to the given event

Example:

The event which gives a sum of ten when two dice are thrown.

Sample space =  $\{ (1,1), (1,2), \dots, (1,6), (2,1), (2,2), \dots, (6,1), \dots, (6,6) \}$

Event space =  $\{ (4,6), (5,5), (6,4) \}$

# Mutually Exclusive Events

When two or more events (call them "A" and "B") are Mutually Exclusive it is **impossible** for them to happen together

Example: Getting value 2 and 5 if a die is thrown

## .....Mutually Exclusive Events

Example:2 :

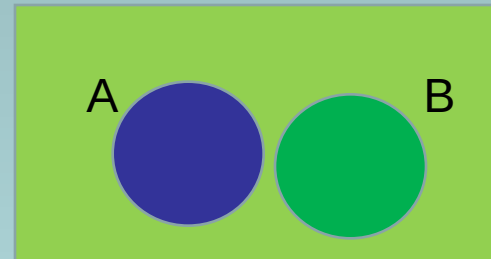
A : Getting an odd number when a die is thrown

B : Getting an even number when a die is thrown

Therefore A & B are Mutually Exclusive Events

But Getting an odd number event and getting number 5 are not mutually exclusive.

Mutually exclusive events shown by using Venn diagram



## ..... Mutually Exclusive Events

### Addition Rule for Mutually Exclusive Events

If A and B are two mutually exclusive events  
 $P(A \text{ or } B) = P(A) + P(B)$

If  $A_1, A_2, \dots, A_n$  are n mutually exclusive events  
 $P(A_1 \text{ or } A_2 \text{ or } \dots \text{ or } A_n) = P(A_1) + P(A_2) + \dots + P(A_n)$

If A and B are not mutually exclusive events  
 $P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$

## Independent events

### Definition

Two events, A and B, are **independent** if the fact that A occurs does not affect the probability of B occurring.

Example:1 :

**P** – A student getting an A pass from a test

**Q** – Another student getting an A pass from the test

## **Multiplication Rule:**

When two events, A and B, are independent, the probability of both occurring is:

$$**P(A \text{ and } B) = P(A) \cdot P(B)**$$

If events,  $A_1, A_2, \dots, A_n$  are independent each other the probability of all occurring together is:

$$**P(A_1 \text{ and } A_2 \text{ and } \dots \text{ and } A_n) = P(A_1)P(A_2)\dots\dots P(A_n)**$$

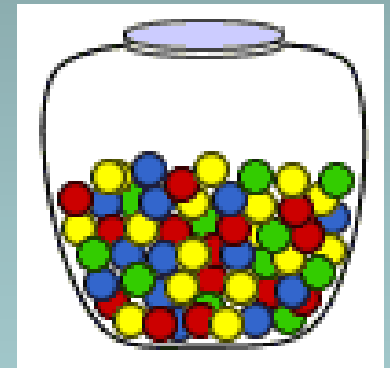
### Example 1:

A jar contains 3 red, 5 green, 2 blue and 6 yellow marbles. A marble is chosen at random from the jar. After replacing it, a second marble is chosen. What is the probability of choosing a green and then a yellow marble?

$$P(\text{green}) = \frac{5}{16}$$

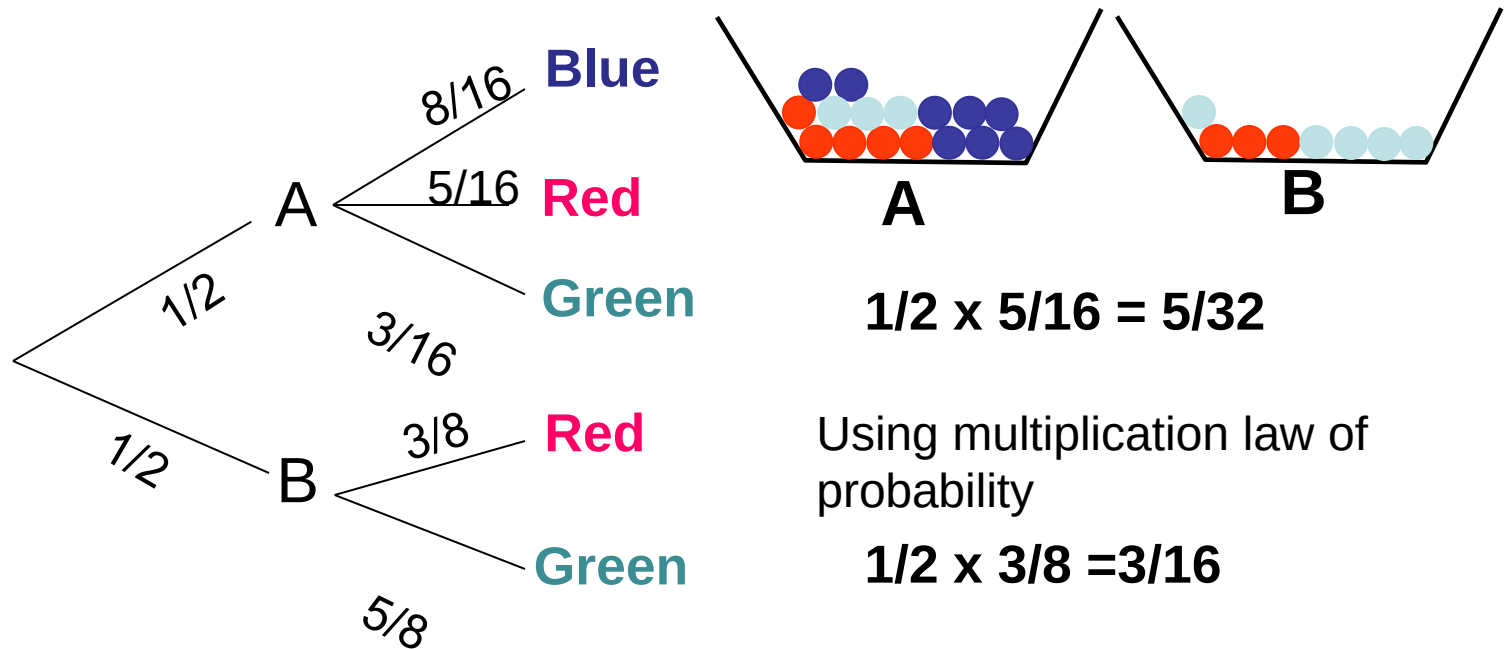
$$P(\text{yellow}) = \frac{6}{16}$$

$$P(\text{green \& yellow}) = \frac{5}{16} \cdot \frac{6}{16}$$



### Example2:

Urn A contains 5 red marbles, 3 green marbles and 8 blue marbles. Urn B contains 3 red marbles and 5 white marbles. An urn is randomly chosen and a marble is selected randomly. Find the probability that a red marble is chosen.



## Example

A student expects that she will obtain grade A passes for subjects statistics and mathematics with probabilities 90% and 70% respectively. Consider that obtaining A passes are independent. Find the probabilities that she will obtain,

- (i) A passes in both subjects,
- (ii) at least in one subject.

## Solution

i) A - obtaining an A pass in Statistics

B - obtaining an A pass in Mathematics

$$P(A) = 0.9 \quad P(B) = 0.7$$

obtaining A passes in both subjects =  $P(A \cap B)$

since events are independent,  $P(A \cap B) = P(A) \cdot P(B)$

$$= 0.9 \times 0.7 = \underline{0.63}$$

ii) A' - not obtaining a A pass in subject Statistics

B' - not obtaining a A pass in subject mathematics

$$P(A') = 1 - 0.9 = 0.1$$

$$P(B') = 1 - 0.7 = 0.3$$

The probability that obtaining A pass at least in one subject =  $1 - P(A' \cap B')$

$$= 1 - 0.03 = \underline{0.97}$$

# Conditional Probability

The **conditional probability** of event A given event B,

$$\text{denoted by } P(A|B) = \frac{P(A \text{ and } B)}{P(B)}$$

provided  $P(B) > 0$

|

### Example 1:

Find the probability that a randomly selected employee being a married male employee given that he is a male employee for following information.

|                         | Male employees, B | Female employees B' |     |
|-------------------------|-------------------|---------------------|-----|
| Married employees, A    | 20                | 35                  | 55  |
| Unmarried employees, A' | 15                | 30                  | 40  |
|                         | 35                | 65                  | 100 |

### Solution

$$P(\text{be an employee Married and Male}) = P(A \cap B) = \frac{20}{100}$$

$$P(\text{be a Male employee}) = P(B) = \frac{35}{100}$$

$$P(\text{given that Male employee be a Married employee}) = P(A|B)$$

$$\text{Therefore } P(A|B) = \frac{P(A \cap B)}{P(B)} = \frac{\frac{20}{100}}{\frac{35}{100}} = \frac{20}{35}$$

|                         | Male employees, B | Female employees B' |     |
|-------------------------|-------------------|---------------------|-----|
| Married employees, A    | 20                | 35                  | 60  |
| Unmarried employees, A' | 15                | 30                  | 40  |
|                         | 35                | 65                  | 100 |

$$n(A \cap B) = 20$$

$$n(B) = 35$$

P(given that Male employee be a Married employee)

$$= \frac{20}{35}$$

### Example 2:

A flight has a 0.9 probability of departure on time from Colombo airport and a 0.8 probability of arrival on time to Singapore airport. The probability a flight which depart on time from Colombo and arrival on time to Singapore is 0.75. Find

- (i) The probability of arrival which was depart on time from Colombo
- (ii) The probability of depart which was arrived on time to Singapore

## Solution

D – Depart from Colombo on time

A – Arrival at on time to Singapore

$$P(D) = 0.9$$

$$P(A) = 0.8$$

$$P(A \cap B) = 0.75$$

(i)  $A|D$  - The probability of arrival which was depart on time from Colombo

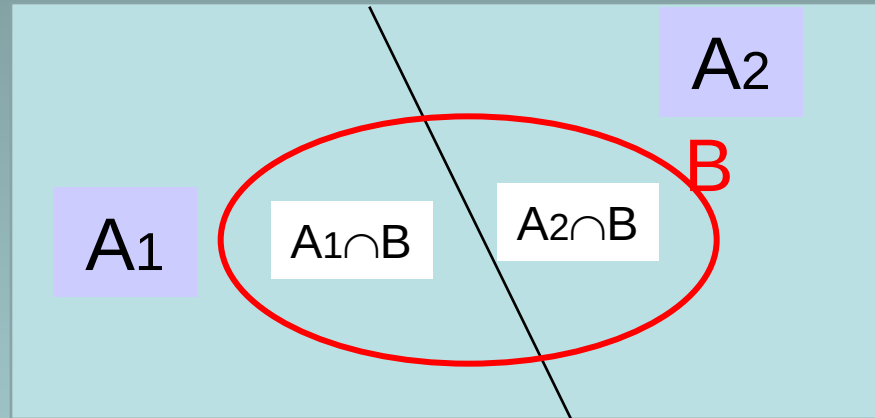
$$P(A|D) = \frac{P(A \cap D)}{P(D)} = \frac{0.75}{0.9} = \frac{5}{6}$$

(ii)  $D|A$  – The probability of depart which was arrived on time to Singapore

$$P(D|A) = \frac{P(A \cap D)}{P(A)} = \frac{0.75}{0.8} = \frac{15}{16}$$

# Baye's Theorem

A partition of the sample space is a collection of disjoint events  $A_1, A_2$  whose union is  $S$ , the sample space.



$A_1 \cap B$  and  $A_2 \cap B$  are mutually exclusive.

$$P(B) = P(A_1 \cap B) + P(A_2 \cap B)$$

$$P(B | A_1) = \frac{P(A_1 \cap B)}{P(A_1)}$$

$$\therefore P(A_1 \cap B) = P(A_1) \cdot P(B | A_1) \quad \therefore P(A_2 \cap B) = P(A_2) \cdot P(B | A_2)$$

$$P(A_1 | B) = \frac{P(A_1 \cap B)}{P(B)}$$

$$P(A_1 | B) = \frac{P(A_1 \cap B)}{P(A_1 \cap B) + P(A_2 \cap B)}$$

$$P(A_1 | B) = \frac{P(A_1) \cdot P(B | A_1)}{P(A_1) \cdot P(B | A_1) + P(A_2) \cdot P(B | A_2)}$$

### Example 1:

A student travels to school by bus or train only. Out of his travels 60% is travelled by bus and 40% is travelled by train. It has found that if he travels by bus the probability of late attendance is 2% and that for by train is 4%.

- (i) Find the probability that he delays attend school.
- (ii) Find the probability that he would have come by bus if it is known that he is late

(i) Let D be the event of delay of the student.

$$\begin{aligned} P(D) &= P(D|Bus)*P(Bus)+ P(D|Train)*P(Train) \\ &= 0.02 \times 0.6 + 0.04 \times 0.4 = \underline{0.028} \end{aligned}$$

(ii) Let D be the event of delay of the student.

$$P(Bus) = 0.6 \quad P(Train) = 0.4$$

$$P(D|Bus) = 0.02 \quad P(D|Train) = 0.04$$

$$P(Bus / D) = \frac{P(D / Bus) * P(Bus)}{P(D / Bus) * P(Bus) + P(D / Train) * P(Train)}$$

$$= \frac{0.02 * 0.6}{0.02 * 0.6 + 0.04 * 0.4}$$

$$= \frac{0.012}{0.028} = \underline{\underline{0.429}}$$

### Example 2:

- Three machines A,B, and C produce, respectively, 40 percent, 10 percent, and 50 percent of the items in a factory. The percentage of defective items produced by the machines is, respectively, 2 percent, 3 percent, and 4 percent. An item from the factory is selected at random.
  - Find the probability that the item is defective.
  - If item is defective , find the probability that the item was produced by:
    - (i) machine A, (ii) machine B, (iii) machine C.

# Probability distributions

a **probability distribution** is a representation of random variable values of occurrence of different possible outcomes in an experiment and their associated probabilities.

## Example 1

Random variable  $X$  is used to denote the Number of getting Heads of a fair coin tossing once ("the experiment"), then the probability distribution of  $X$  would take the value  $1/2$  for  $X = 1$  and  $x=0$ .

| $X$    | 0     | 1     |
|--------|-------|-------|
| $P(x)$ | $1/2$ | $1/2$ |

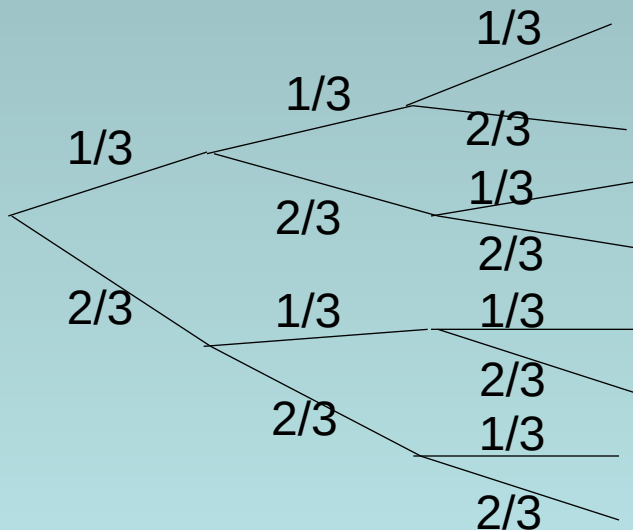
## Example 2

Random variable  $X$  is used to denote the value of the outcome getting of a fair die tossing once.

| $X$    | 1   | 2   | 3   | 4   | 5   | 6   |
|--------|-----|-----|-----|-----|-----|-----|
| $P(x)$ | 1/6 | 1/6 | 1/6 | 1/6 | 1/6 | 1/6 |

## Example 3

Random variable  $X$  is used to denote the number of heads of getting of a biased coin tossing thrice with probability of getting Head is  $2/3$ .



| $X$    | 0    | 1   | 2   | 3    |
|--------|------|-----|-----|------|
| $P(x)$ | 1/27 | 2/9 | 4/9 | 8/27 |

For any probability distribution  $\Sigma p(x) = 1$

Expectation

$$E(x) = \Sigma xp(x)$$

Example 1

| X    | 1   | 2   | 3   | 4   | 5   | 6   |
|------|-----|-----|-----|-----|-----|-----|
| P(x) | 1/6 | 1/6 | 1/6 | 1/6 | 1/6 | 1/6 |

$$E(x) = \Sigma xp(x)$$

$$E(x) = (1)\left(\frac{1}{6}\right) + (2)\left(\frac{1}{6}\right) + (3)\left(\frac{1}{6}\right) + (4)\left(\frac{1}{6}\right) + (5)\left(\frac{1}{6}\right) + (6)\left(\frac{1}{6}\right)$$

$$E(x) = \underline{\underline{3.5}}$$

Expectation is the mean of the probability distribution.

$$\mu = E(x)$$

# Variance

$$V(x) = E(x^2) - [E(x)]^2$$

Example 1

| X    | 1   | 2   | 3   | 4   | 5   | 6   |
|------|-----|-----|-----|-----|-----|-----|
| P(x) | 1/6 | 1/6 | 1/6 | 1/6 | 1/6 | 1/6 |

$$E(x) = 3.5$$

$$E(x^2) = (1^2)\left(\frac{1}{6}\right) + (2^2)\left(\frac{1}{6}\right) + (3^2)\left(\frac{1}{6}\right) + (4^2)\left(\frac{1}{6}\right) + (5^2)\left(\frac{1}{6}\right) + (6^2)\left(\frac{1}{6}\right)$$

$$E(x^2) = (1 + 4 + 9 + 16 + 25 + 36)\left(\frac{1}{6}\right) \quad E(x^2) = \left(\frac{91}{6}\right)$$

$$V(x) = \frac{91}{6} - \left(\frac{7}{2}\right)^2 = 15.16 - 12.25 = \underline{\underline{2.92}}$$

Standard deviation of the probability distribution

$$\sigma(x) = \sqrt{V(x)}$$

Find the probability distribution of the random variable  $X$  where  $X$  is the number of winnings of a game with 3 rounds with a probability of win is  $2/3$ . Calculate the expectation and standard deviation.

Solution

| $X$    | 0      | 1     | 2     | 3      |
|--------|--------|-------|-------|--------|
| $P(x)$ | $1/27$ | $2/9$ | $4/9$ | $8/27$ |

$$E(x) = 0 \times \frac{1}{27} + 1 \times \frac{2}{9} + 2 \times \frac{4}{9} + 3 \times \frac{8}{27} = \frac{54}{27} = \underline{\underline{2}}$$

$$E(x^2) = 0 \times \frac{1}{27} + 1 \times \frac{2}{9} + 4 \times \frac{4}{9} + 9 \times \frac{8}{27}$$

$$E(x^2) = 4.67$$

$$\sigma(x) = \sqrt{4.67 - 4} = \underline{\underline{1.63}}$$

END